AI AGENTS HACKATHON · AMERICAN UNIVERSITY IN DUBAI

Zill & Shams

ظل وشمس *“shade and sun”*

Shade when you need it. Sun when you want it.

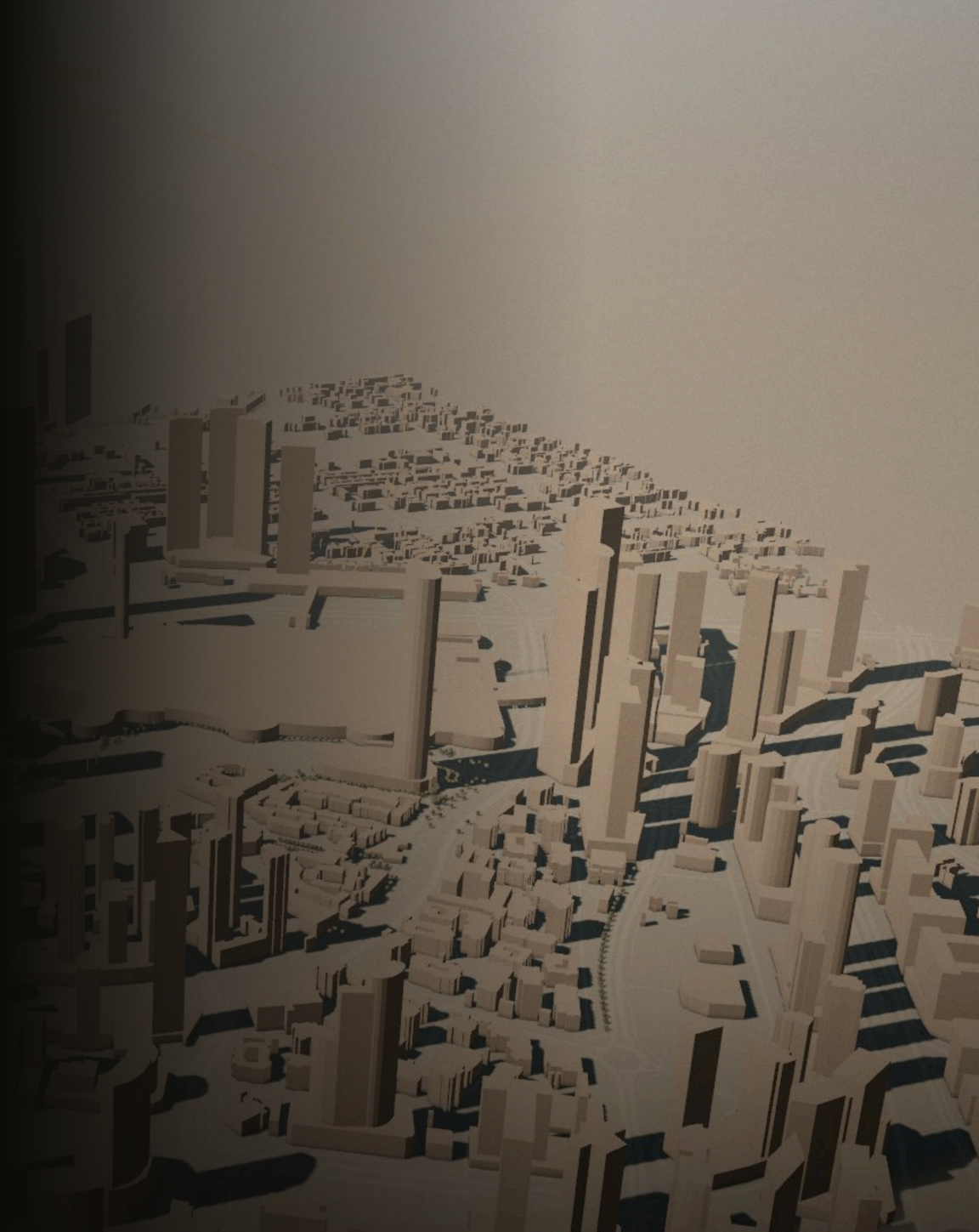
An AI agent that plans every walk and ride around the sun.

Maps plan around time. Nobody plans around the sun.

01 In hot cities, the route you pick decides how much direct sun you take, and in summer that is a health risk.

02 In dark winters, people get too little daylight. Oslo has about 6 hours of daylight in late December.

03 The sun and shade move every minute, so a static “shady street” or “sunny street” list is never right for long.



03 · WHY IT MATTERS

72%

of 12,346 adults in an Abu Dhabi study were vitamin D deficient, and another 10% insufficient, in one of the sunniest places on earth.

BMC RESEARCH NOTES, 2019

70,000+

delivery riders work across Dubai.

DUBAI RTA

18.72 million

international overnight visitors came to Dubai in 2024.

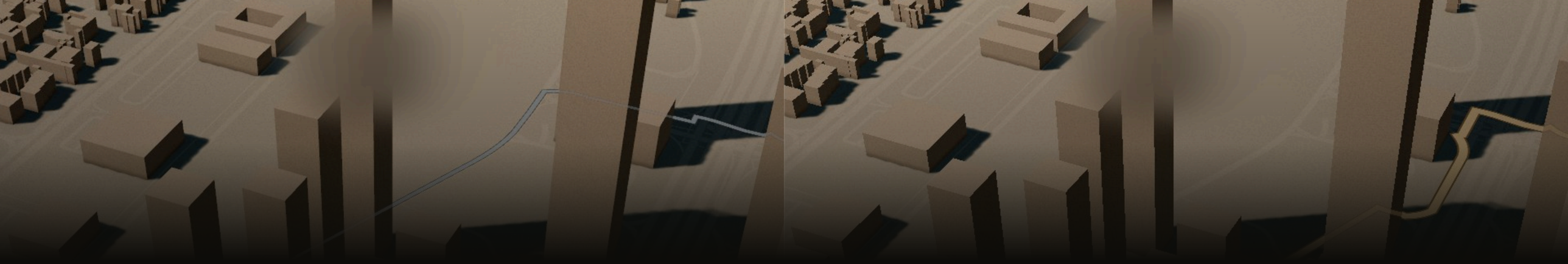
DUBAI DET VIA WAM

12:30–15:00

The UAE bans outdoor work in direct sun, 15 June to 15 September, for the 21st year in 2025.

UAE MINISTRY OF HUMAN RESOURCES AND EMIRATISATION

Sources: BMC Research Notes 2019, vitamin D in Abu Dhabi (n=12,346): ncbi.nlm.nih.gov/pmc/articles/PMC6693241 · Dubai RTA, delivery rider programmes: rta.ae (news: 23 temporary rest areas to protect delivery riders from heat stress) · Dubai DET via WAM: wam.ae/article/bi46kc8-dubai-welcomes-1872-million-international-visitors · MoHRE, Midday Break 2025: mohre.gov.ae/en/media-center/news/3/6/2025 · Oslo daylight at the December solstice (about 5 h 54 min): fjordtours.com/en/norway/people-and-culture/facts/winter-solstice



04 · THE PRODUCT

One engine, two modes.

SHADE MODE · FOR HOT CITIES

Finds the coolest route: the most shade and the least direct sun, using air-conditioned links such as the Dubai Metro to Dubai Mall walkway.

SUN MODE · FOR DAYLIGHT AND WELL-BEING

Finds the sunniest route for people who want more daylight: winter cities, office workers, anyone short on sun.

Same engine. The preference flips.

How it works, in one picture.



01 Open map data

→

02 3D model of the city

→

03 Where the sun and shadows fall, every 10 minutes

→

04 The best route for you

→

05 An AI agent that plans, checks and adjusts

Works for any city with OpenStreetMap data. Our first city: Downtown Dubai (8,602 buildings, 7,659 street segments, 744 trees).



06 · PROOF IT IS REAL

A real walk computed by the engine: Burj Park → Business Bay Metro, a summer afternoon at 15:30.

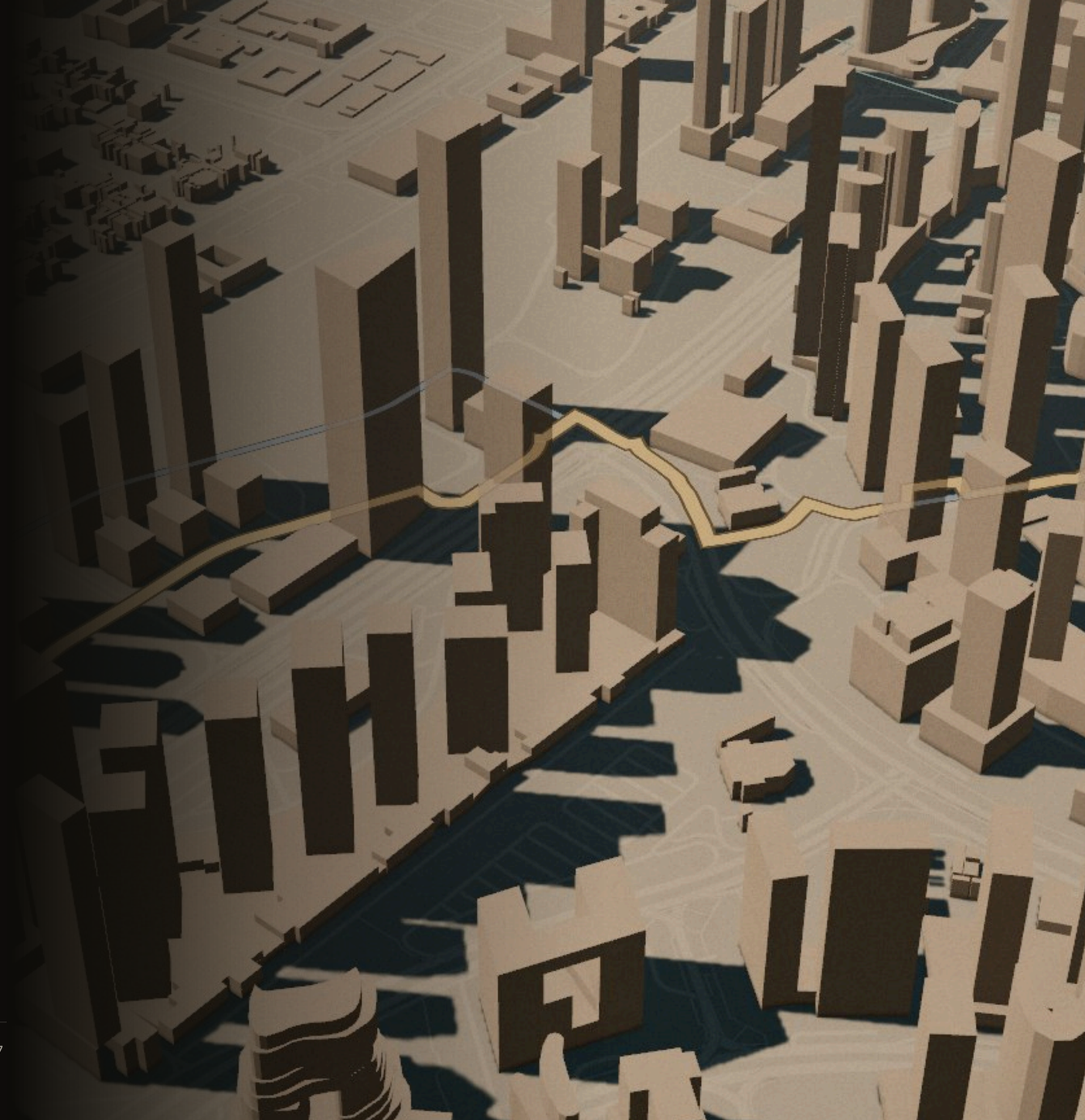
6.2	40%	2.2
MINUTES LESS DIRECT SUN	LESS SUN	ONLY MINUTES LONGER

Computed from real map data, not a mock-up. Live at zill.itshamza.me

Model estimate; shade from buildings and mapped trees. Awnings and wind are not modelled yet.

An agent, not just a map.

PLANS	Understands “I’m walking to the metro, I tire easily in the heat” and compares options.
CHECKS	A second AI step verifies every claim; every number comes from the engine, never made up.
ADAPTS	If you are delayed or a street is closed, it replans and asks before changing your plan.



Who pays.

SEGMENT	WHAT THEY GET	HOW THEY PAY
Delivery fleets targets such as Talabat, Deliveroo, Careem	Heat-aware routes and shift pacing for riders	Per rider per month
Hotels and tourism boards	“Walk there in the shade, or in the sun” guest concierge, so guests walk short distances instead of booking a cab	Per hotel per month
Employers of outdoor workers	Shift planner that respects the midday break	Per site per month
Corporate well-being and insurers	Daylight-walk programmes for staff	Per employee per year
Cities with dark winters Nordics, Canada	Sunniest-route mode for residents	City licence

Free for individuals, which builds reach and data. Prices are hypotheses we are validating.

09 · WHO IT IS FOR

Four people, one product.

SHADE

Rider in Dubai

JULY 14:00

The coolest route, plus a rest stop in air conditioning.

SHADE

Hotel guest

BURJ PARK

A 25-minute walk in the shade instead of a cab.

SUN

Office worker

DUBAI, WINTER

A 15-minute sunny walk at lunch for daylight.

SUN

Resident of Oslo

DECEMBER

The sunniest path during the few daylight hours.

Sun mode is about daylight and well-being, not medical advice.

Business model.

Low cost to add a city: it is built from open data automatically.

01 B2B software subscriptions (fleets, hotels, employers) as the main revenue.

02 A pay-per-route API for delivery and mobility apps.

03 City and tourism-board licences.

04 A free consumer app for reach.

Go-to-market and roadmap.

01 · DUBAI

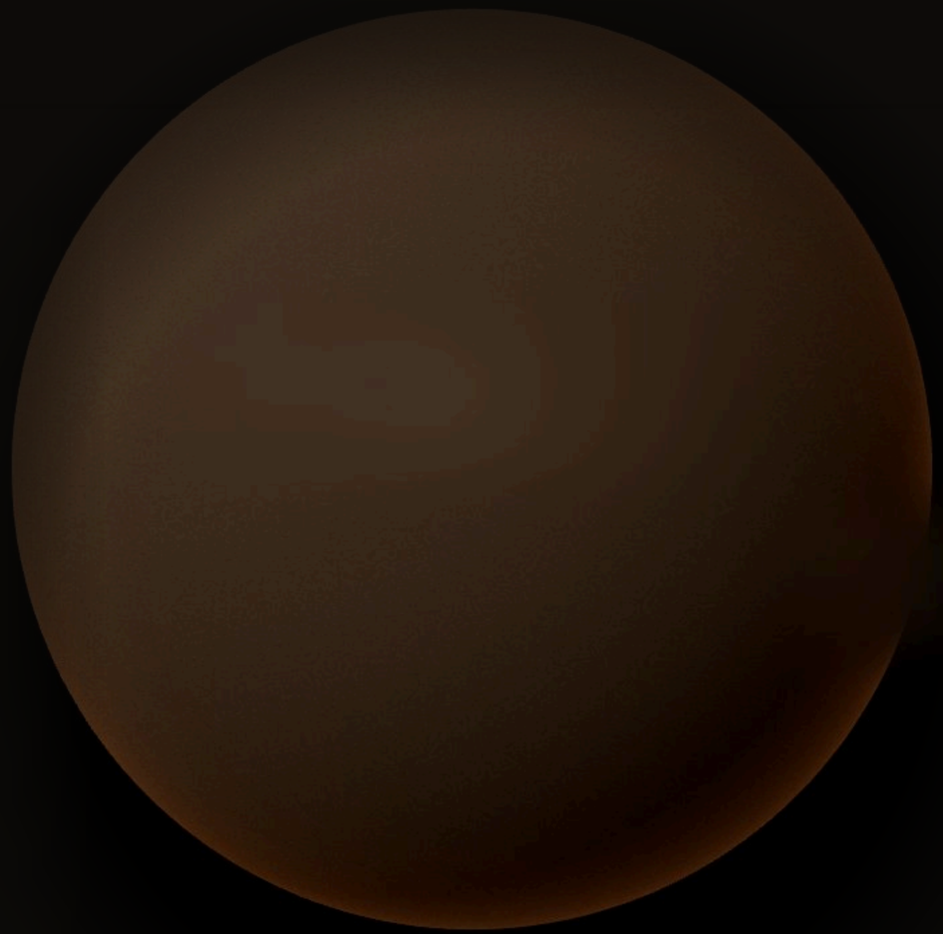
A pilot with one delivery fleet and one hotel. The prototype is already live.

02 · UAE AND GULF

Abu Dhabi, Doha, Riyadh; the sun mode for winter well-being.

03 · NORTHERN CITIES

A winter daylight pilot in a city like Oslo or Toronto.



What already works.

01 A live prototype at zill.itshamza.me: 3D city, shade engine, routing and an AI agent.

02 Real data for Downtown Dubai and real sun calculations; 296 automated tests pass.

03 Honest limits: shade comes from buildings and mapped trees; field measurements are the next step.



Zill & Shams

ظل وشمس

Shade when you need it. Sun when you want it.

ABHAMZAY: MOHAMMED HAMZA ANSARI · ABHAY PREETHAN

TRY IT: ZILL.ITSHAMZA.ME